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|-----------|-----------------------------------|
| Status    | Finished                          |
| Started   | Monday, 23 December 2024, 5:33 PM |
| Completed | Friday, 6 December 2024, 2:48 PM  |
| Duration  | 17 days 2 hours                   |

Question **1**

Correct

Marked out of  
3.00

🚩 Flag question

The k-digit number N is an Armstrong number if and only if the k-th power of each digit sums to N.

Given a positive integer N, return true if and only if it is an Armstrong number.

Example 1:

Input:

153

Output:

true

Explanation:

153 is a 3-digit number, and  $153 = 1^3 + 5^3 + 3^3$ .

Example 2:

Input:

123

Output:

false

Explanation:

123 is a 3-digit number, and  $123 \neq 1^3 + 2^3 + 3^3 = 36$ .

Example 3:

Input:

1634

Output:

true

Note:

$1 \leq N \leq 10^8$

**Answer:** (penalty regime: 0 %)

```
1 #include<stdio.h>
2 #include<math.h>
3 int main()
4 {
5     int n;
6     scanf("%d",&n);
7     int x=0,n2=n;
8     while(n2!=0)
9     {
10         x++;
11         n2=n2/10;
12     }
13     int sum=0;
14     int n3=n,n4;
```

```

15 while(n3!=0)
16 {
17     n4=n3%10;
18     sum=sum+pow(n4,x);
19     n3=n3/10;
20 }
21 if(n==sum)
22 {
23     printf("true");
24 }
25 else
26 {
27     printf("false");
28 }
29 return 0;
30 }

```

|   | Input | Expected | Got   |   |
|---|-------|----------|-------|---|
| ✓ | 153   | true     | true  | ✓ |
| ✓ | 123   | false    | false | ✓ |

Passed all tests! ✓

Question 2

Correct

Marked out of 5.00

Flag question

Take a number, reverse it and add it to the original number until the obtained number is a palindrome. Constraints  $1 \leq \text{num} \leq 99999999$  Sample Input 1 32 Sample Output 1 55 Sample Input 2 789 Sample Output 2 66066

**Answer:** (penalty regime: 0 %)

```

1 #include<stdio.h>
2 int main()
3 {
4     int rn,n,nt=0,i=0;
5     scanf("%d",&n);
6     do
7     {
8         nt=n;rn=0;
9         while(n!=0)
10        {
11            rn=rn*10+n%10;
12            n=n/10;

```

```

13     }
14     n=nt+rn;
15     i++;
16 }
17 while(rn!=nt||i==1);
18 printf("%d",rn);
19 return 0;
20 }

```

|   | Input | Expected | Got   |   |
|---|-------|----------|-------|---|
| ✓ | 32    | 55       | 55    | ✓ |
| ✓ | 789   | 66066    | 66066 | ✓ |

Passed all tests! ✓

Question **3**

Correct

Marked out of  
7.00

🚩 Flag question

A number is considered lucky if it contains either 3 or 4 or 3 and 4 both in it. Write a program to print the nth lucky number. Example, 1st lucky number is 3, and 2nd lucky number is 4 and 3rd lucky number is 33 and 4th lucky number is 34 and so on. Note that 13, 40 etc., are not lucky as they have other numbers in it.

The program should accept a number 'n' as input and display the nth lucky number as output.

Sample Input 1:

3

Sample Output 1:

33

Explanation:

Sample Input 2:

34

Sample Output 2:

33344

**Answer:** (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n=1,i=0,nt,co=0,e;
5     scanf("%d",&e);
6     while(i<e)
7     {
8         nt=n;
9         while(nt!=0)
10        {
11            co=0;
12            if(nt%10!=3&&nt%10!=4)
13            {
14                co=1;
15                break;
16            }
17            nt=nt/10;
18        }
19        if(co==0)
20        {
21            i++;
22        }
23        n++;
24    }
25    printf("%d",--n);
26    return 0;
27 }
```

|   | Input | Expected | Got   |   |
|---|-------|----------|-------|---|
| ✓ | 34    | 33344    | 33344 | ✓ |

Passed all tests! ✓

Finish review